IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Data Collection via API
  - Data Collection via Web Scraping
  - Data Wrangling
  - EDA with SQL
  - EDA with Visualizations
  - Interactive Visual Analytics with Folium
  - Interactive Dashboard with Plotly Dash
  - Predictive Analysis with Machine Learning

- Summary of all results
  - Landing outcomes improved over time
  - The orbits ES-L1, GEO, HEO, and SSO had highest success rates.
  - Heavier payloads had higher failure rates earlier on but improved over time
  - The launch site KSC-LA had the highest success rate.
  - Decision Tree Classifier was the most accurate in predicting landing outcomes

Introduction

- Project background and context:
  - SpaceX, an aerospace manufacturer, launches rockets to the International Space Station. However, they are able to do it at a relatively inexpensive price compared to other space companies. This is because they advertise Falcon 9 rocket launches at a price of 62 million dollars, whereas other companies cost upwards of 165 million dollars. One reason SpaceX can advertise at a much cheaper price is because they re-use their first stage of the launch, assuming it lands. Therefore, if we, the company SpaceY who wants to compete with SpaceX, can determine if the first stage will land, we can predict the financial risk. determine the cost of a launch for our company, and compete with SpaceX.
- Problems you want to find answers
  - Main Problem: Predicting the successful landing of the Falcon 9's first stage in a SpaceX rocket launch.
    - Sub Problem: What factors most significantly influence the landing outcome?
    - Sub Problem: Can we use a predictive model to accurately calculate the landing outcome?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Requested data via a REST API
  - Web scraping from HTML Table in Wikipedia
- Perform data wrangling
  - Extracted relevant data records, handled missing and null values, and used binary classification to label landing outcomes
- Perform exploratory data analysis (EDA) using visualization and SQL
  - Performed SQL queries to analyze relationships between variables
  - Created visualizations to see trends and correlations within the dataset
- Perform interactive visual analytics using Folium and Plotly Dash
  - Created Folium maps to see locations, landing outcomes, and other geographical markers near launch sites
  - Used Dash to create a dashboard of success rates of all launch sites, individual success of each launch site, and payload range relative to landing outcome of each booster version
- Perform predictive analysis using classification models
  - Built, tuned, and evaluated multiple machine learning models to predict the probability of successful first-stage landings.

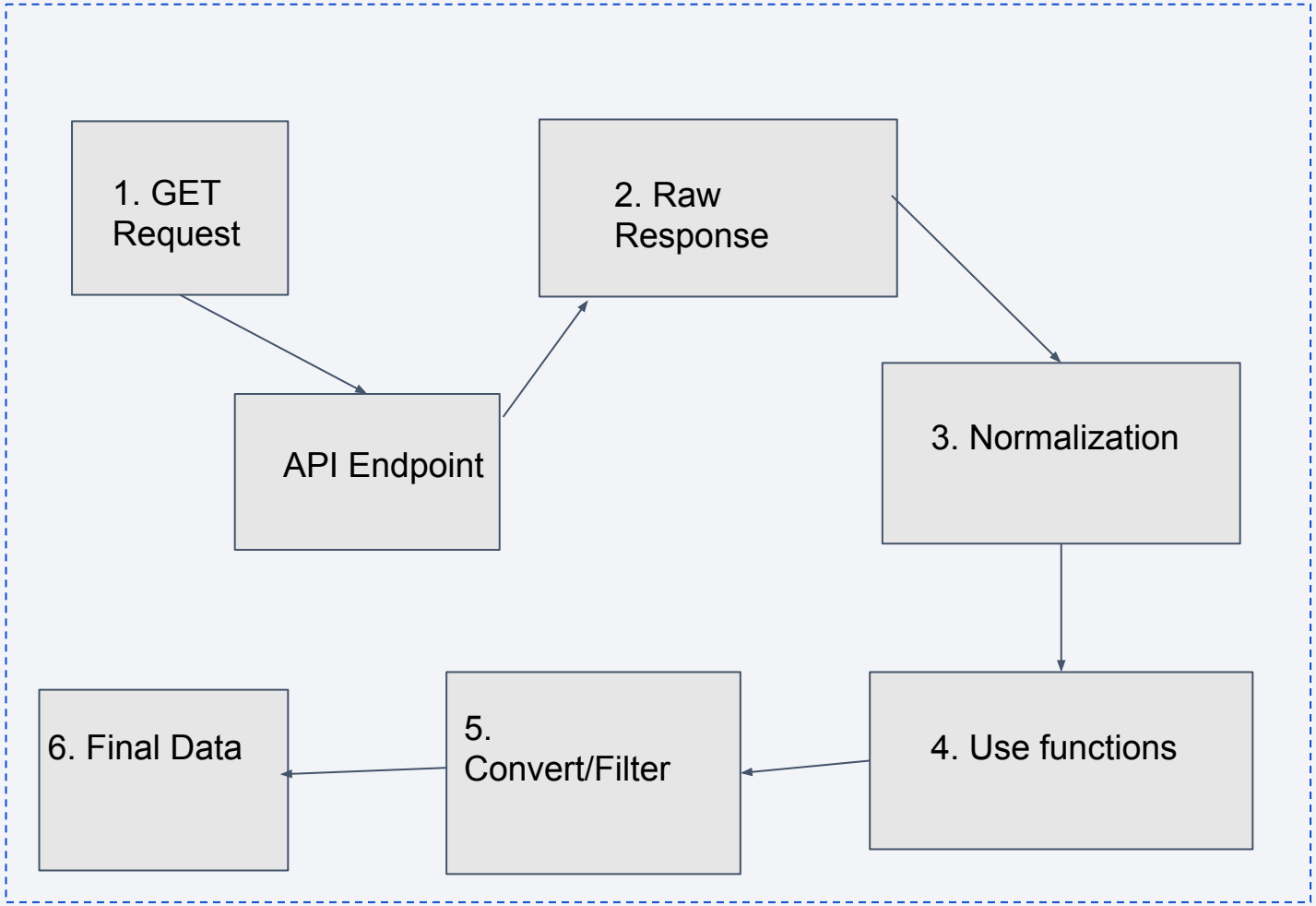
Data Collection

- Data was collected using two retrieval techniques:
  - REST API:
    - Performed a get request on the SpaceX API (v4/launches/past) to retrieve historical launch data
    - Converted raw response into a JSON object
    - Converted JSON object into a Pandas dataframe
    - Created functions (getBoosterVersion, getLaunchSite, etc) to understand the data within the “ID” codes and replace them with actual names
  - Web Scraping HTML Table on Wikipedia
    - Requested Falcon9 launch data from HTML page as an HTML response
    - Instantiated the HTML response as a BeautifulSoup object
    - Extracted relevant data and created a dataframe with it

Data Collection – SpaceX API

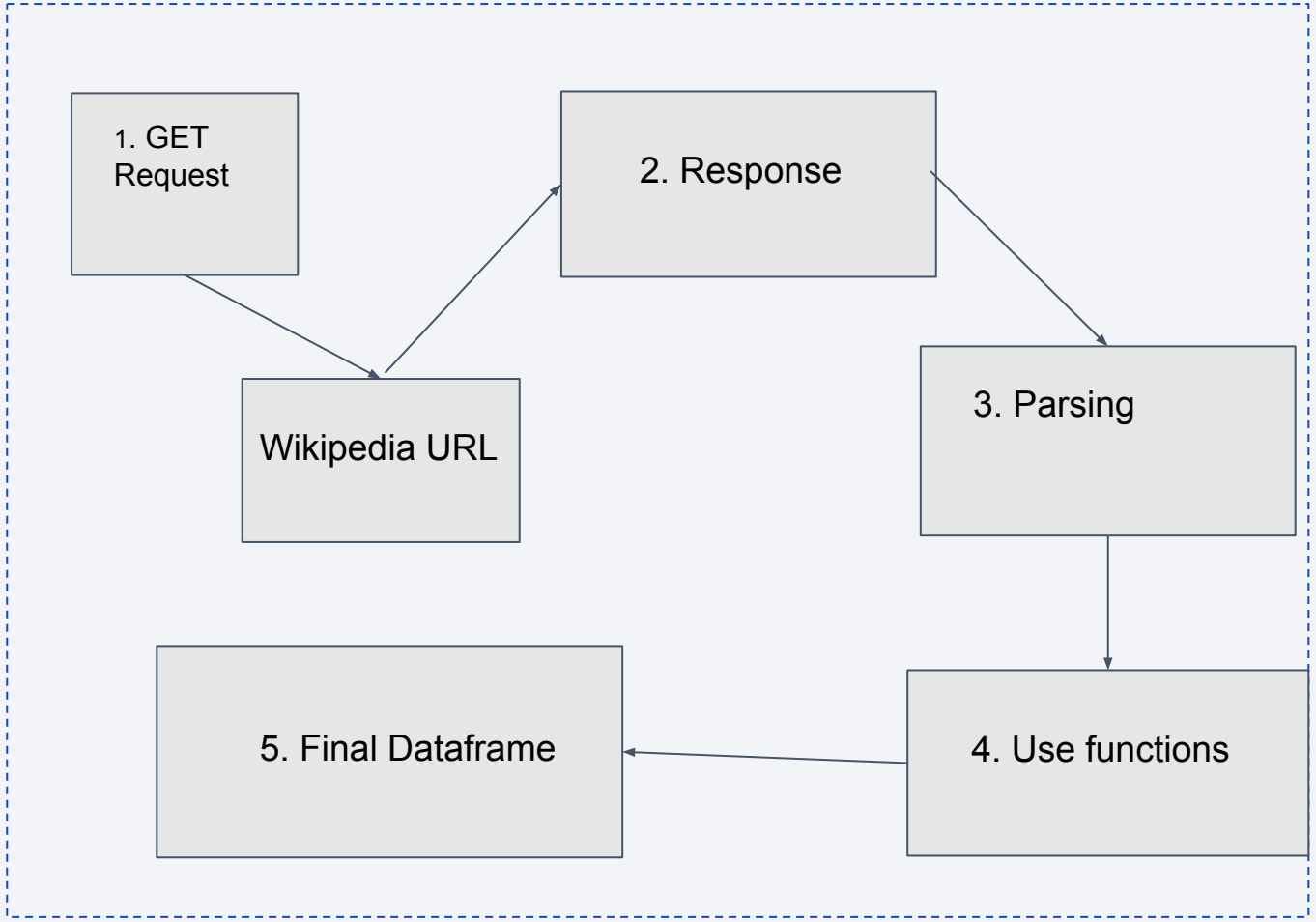
1. Initial data retrieved from API Endpoint using a GET Request.
2. API sends back a JSON response.
3. Normalized JSON response to turn the messy data into readable, understandable data via a Pandas Dataframe (temporary)
4. Use custom functions to find information about ID's and convert ID's to names with a dictionary
5. Convert dictionary to final Pandas Dataframe and filter dataframe to keep only relevant features.
6. Final, useable data

GitHub URL



Data Collection - Scraping

1. HTTP GET Request to Wikipedia URL
2. Returns source code as a string of HTML
3. Convert HTML code to BeautifulSoup object
4. Use functions to find and extract relevant data
5. Store scraped values into a Pandas Dataframe



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Data Wrangling

- Explore Data:
  - Calculated percentages of missing values per attribute
  - Identify numeric and categorical columns
  - Determine launches per site
  - Determine number of orbits and how much they were used AND calculate the mission outcomes of the orbits
- Create labels:
  - Create landing labels for outcomes

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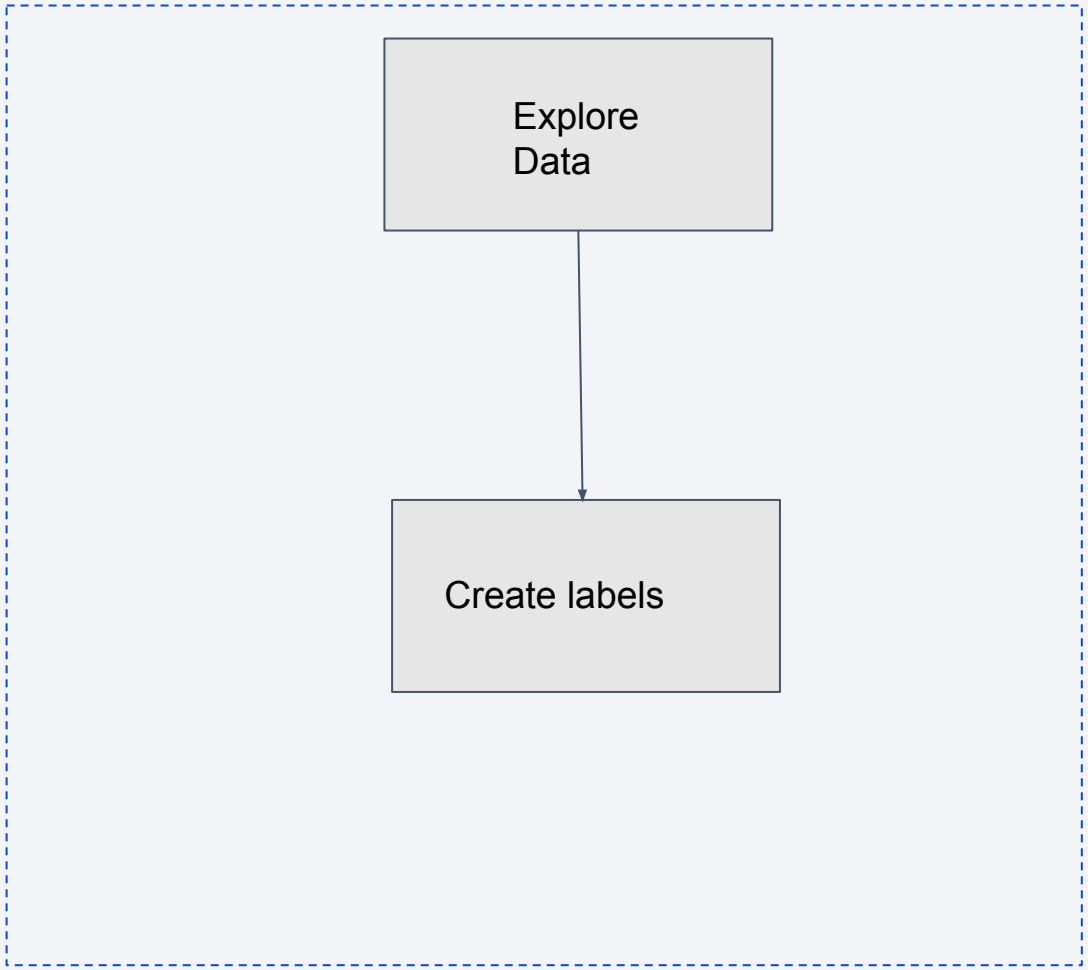

EDA with Data Visualization

- Visualize relationships between variables to see their importance regarding successful landing:
  - Flight Number and Payload Mass
  - Flight Number and Launch Site
  - Payload Mass and Launch Site
  - Success Rate of Orbit Type
  - Flight Number and Orbit Type
  - Payload Mass and Orbit Type
  - Launch Success Yearly Trend
- Create dummy variables for Orbit, Launch Site, Landing Pad, and Serial
- Convert numeric columns to float64

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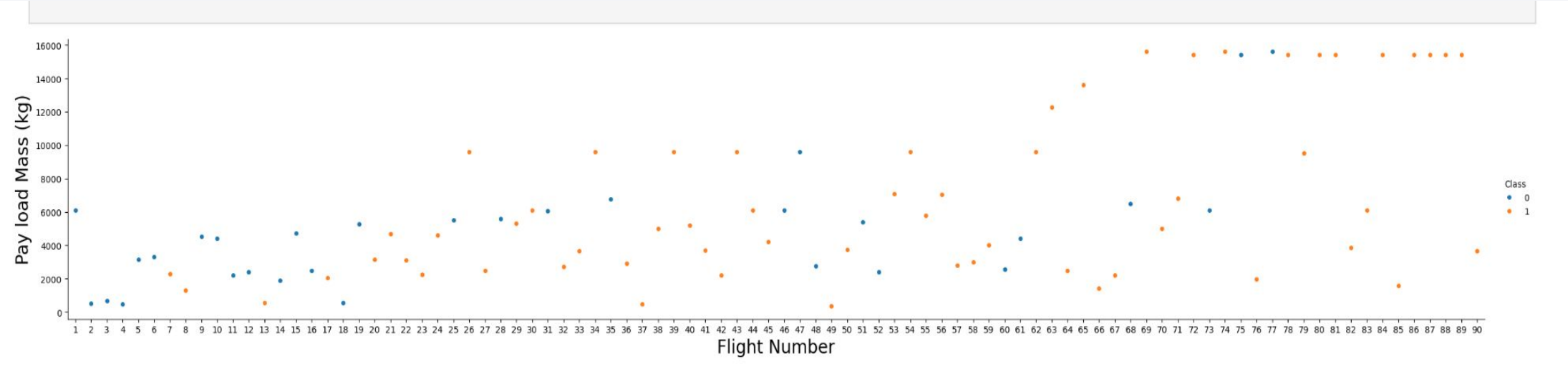
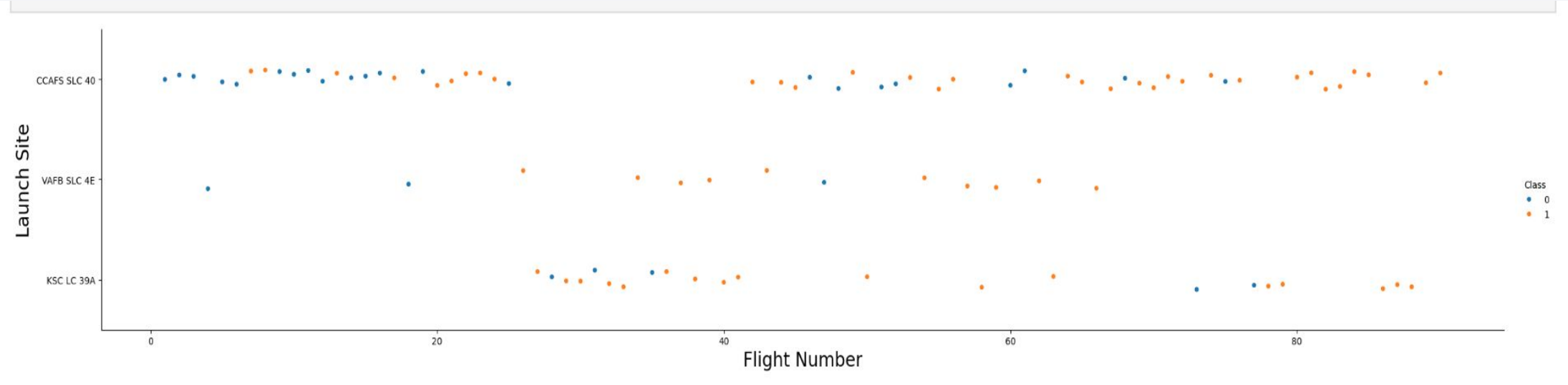
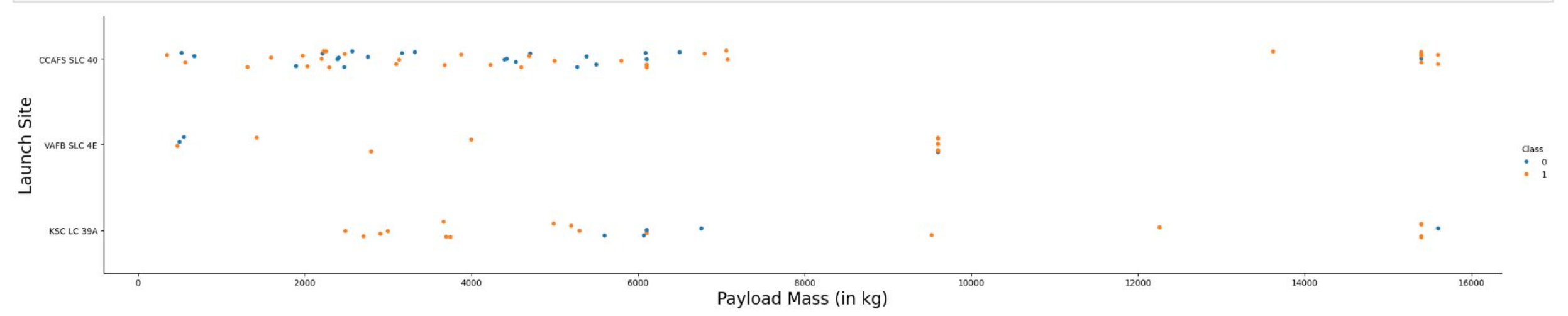
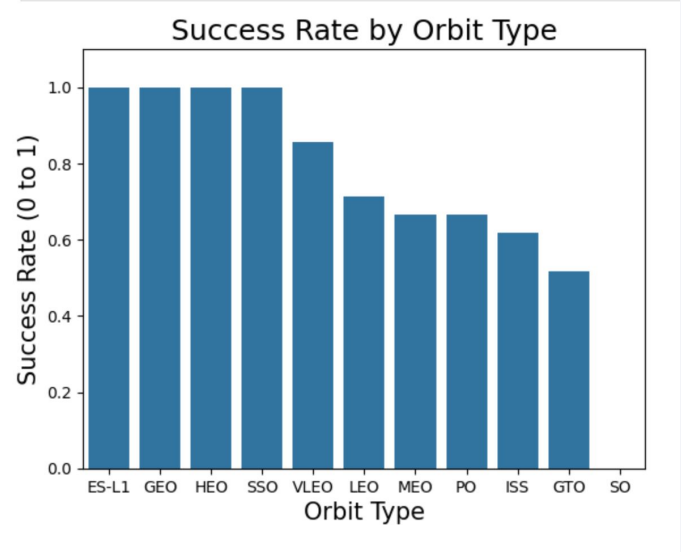
Charts - Data Visualization

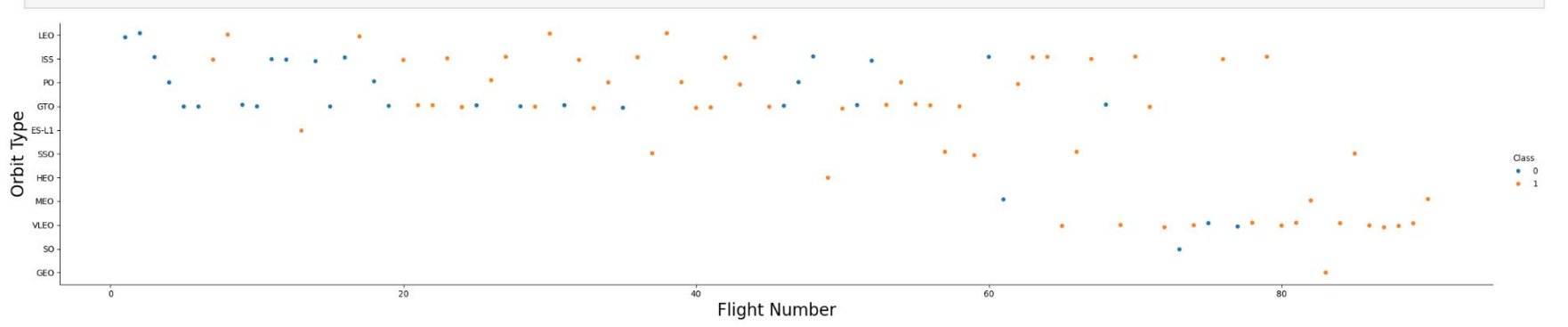




Charts - Data Visualization







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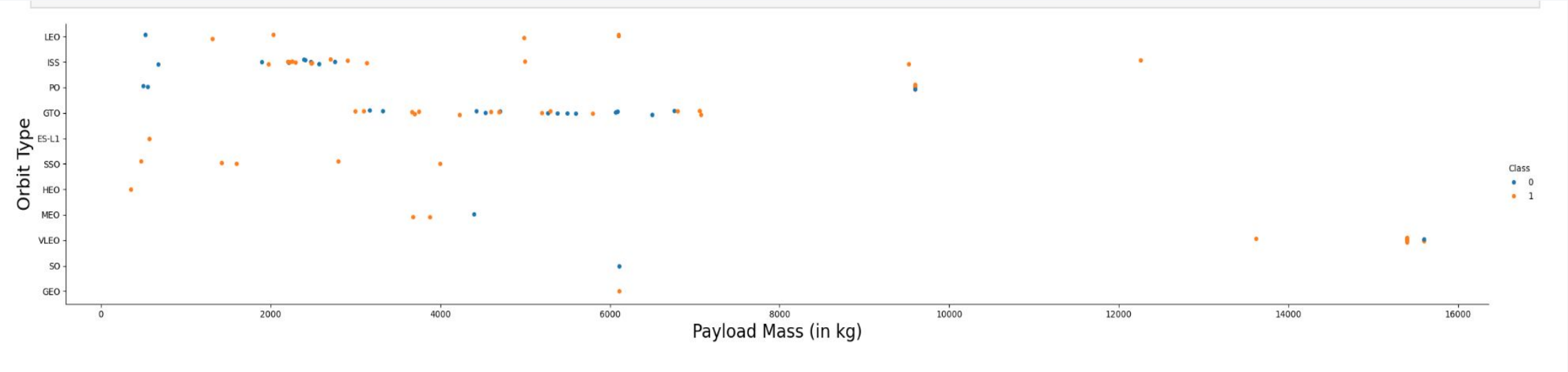

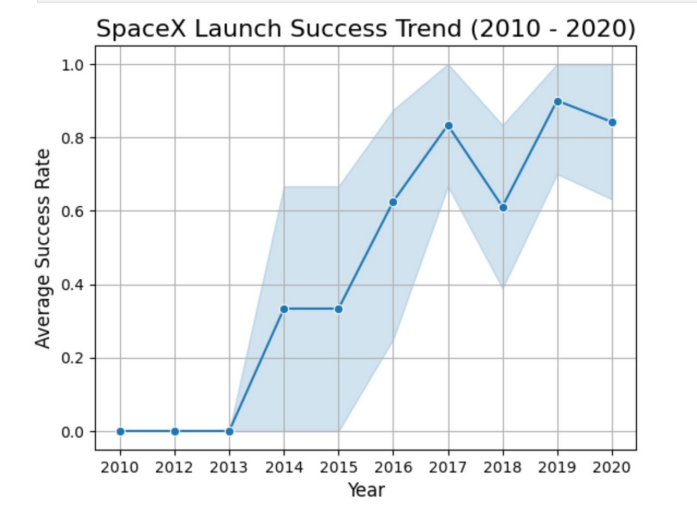

EDA with SQL

- SQL Queries:
  - Unique launch sites
  - 5 records where launch site began with ‘CCA’
  - Total payload mass carried by boosters launched by NASA
  - Average payload mass carried by booster F9 v1.1
  - Date of first successful landing outcome on ground pad
  - Names of boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
  - Total number of successful and failure mission outcomes
  - Booster versions that carried maximum payload mass
  - Records that displayed month names, failure landing outcomes in drone ship, booster versions, and launch sites for the months in 2015
  - Ranked count of landing outcomes between 2010-06-04 and 2017-03-20 in descending order

GitHub URL

Build an Interactive Map with Folium

- To find geological patterns, these map objects were created and added:
  - Launch sites (markers and circles)
  - Launch outcomes for each site (markers and circles)
  - Distance between launch sites and proximities (lines)

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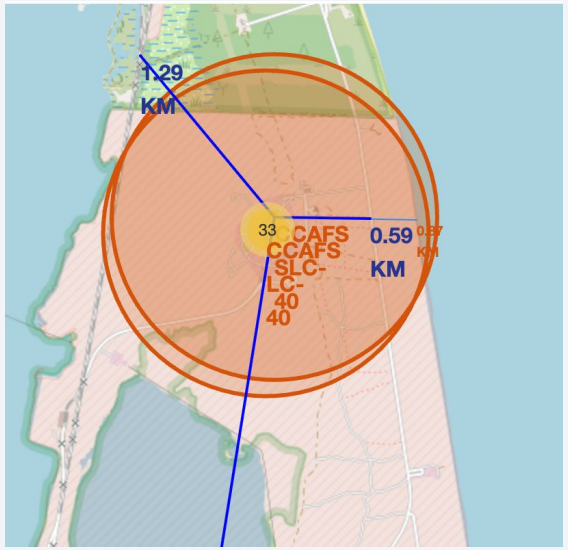

Build a Dashboard with Plotly Dash

- To enable interactive exploration of data, the following was added:
  - Dropdown selector to choose launch site/s
    - Pie Chart displayed based on selection, showing successful launches per launch site/s (note: if one launch site was selected, failure rates were also shown)
    - Scatter plot of payload range and landing outcome grouped by booster version was displayed per launch site/s selected
  - Range slider for selecting payload range for scatter plot

GitHub URL

Predictive Analysis (Classification)

1. Load data
2. Convert Y to numpy array
3. Standardize data in X
4. Split data
5. Use GridSearchCV to find best parameters for:
   a. Logistic Regression
   b. SVC
   c. Decision Tree Classifier
   d. KNN Classifier
6. Calculate accuracy of model on test data (confusion matrix, score method)
7. Compare results and choose best model (Decision Tree Classifier)

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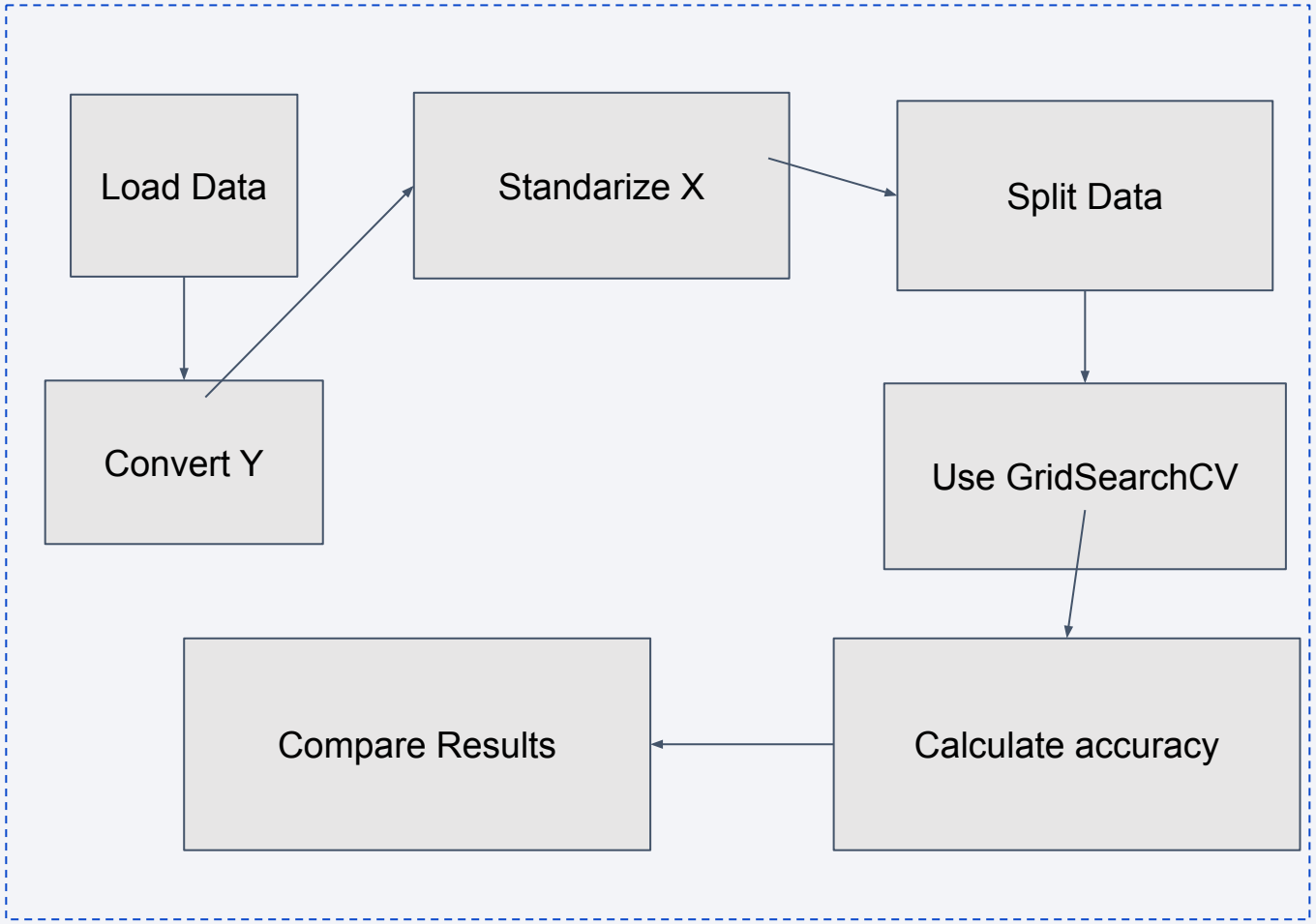


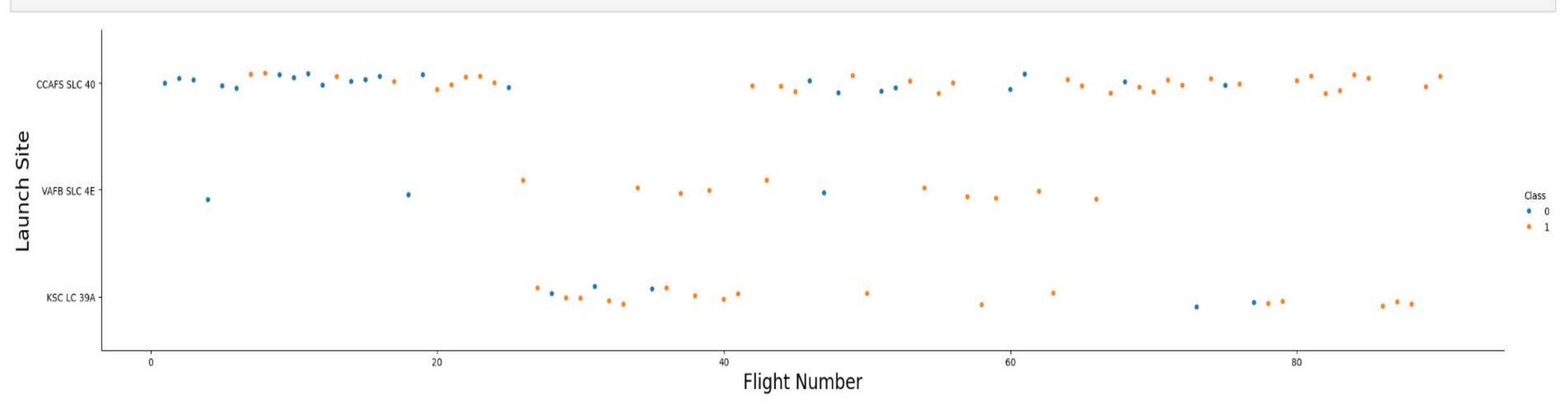
Results

- Exploratory data analysis results
  - Success rates of landing outcomes increased over time, especially after 2015
  - Success rates of landing outcomes are generally higher between paylaod mass of 2000 to 5000
  - As flight number increases, landing outcomes are more successful
  - Orbits ES-L1, GEO, HEO, and SSO have the highest success rates
- Interactive analytics demo in screenshots
  - Launch sites tend to be further from cities but closer to highways, railroads, and oceans
  - KSC launch site had the most successful launches
  - Payload mass between 2000 and 5000 had the most successful launches, with FT booster version having the most
- Predictive analysis results
  - Decision Tree Classifier model was the best model based on score method and confusion matrix

Section 2

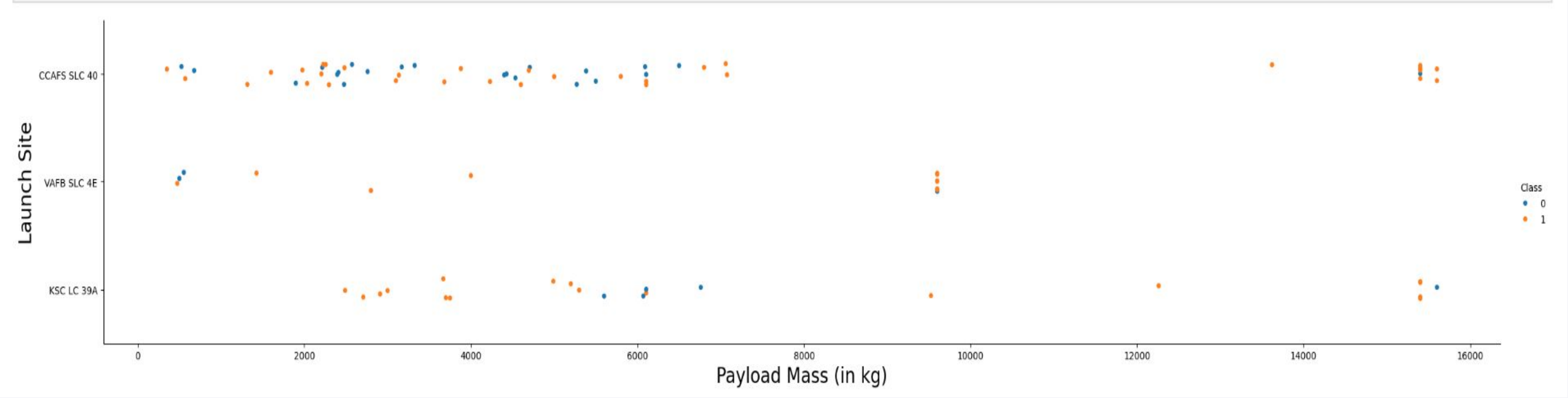
Insights drawn from EDA

Flight Number vs. Launch Site



1. As flight number increases, succesful landing outcomes tend to increase for CCAFS launch site.
2. This pattern is also observed for VAFB launch site, however, more data is needed to confirm this.
3. For KSC launch site, this pattern is not observed. We can see successful landing outcomes regardless of flight number. However, it's important to note that they do not have flight numbers less than 25 (approximately).
4. This suggests that CCAFS was a R&D phase while KSC was an operational maturity phase.

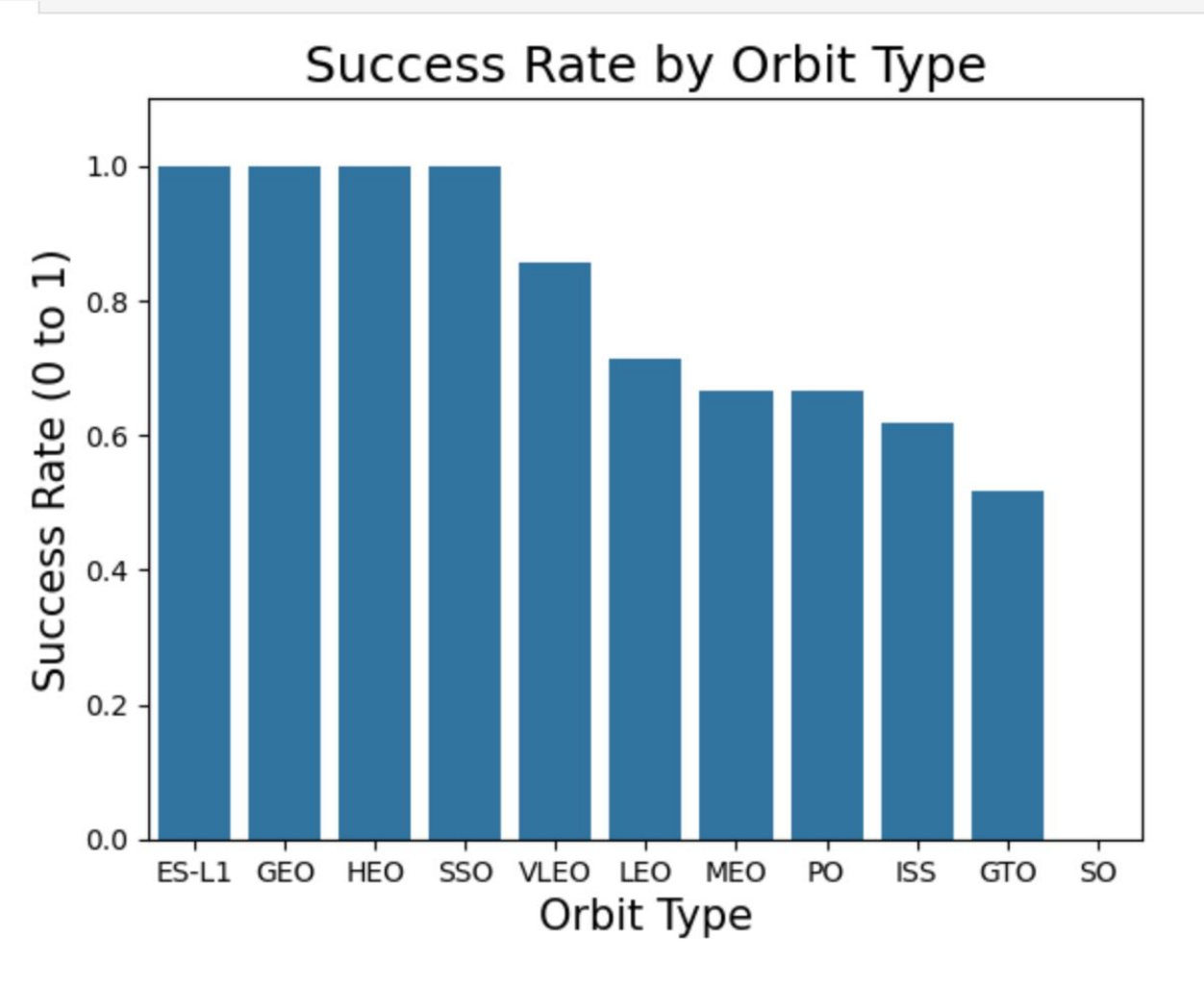
Payload vs. Launch Site



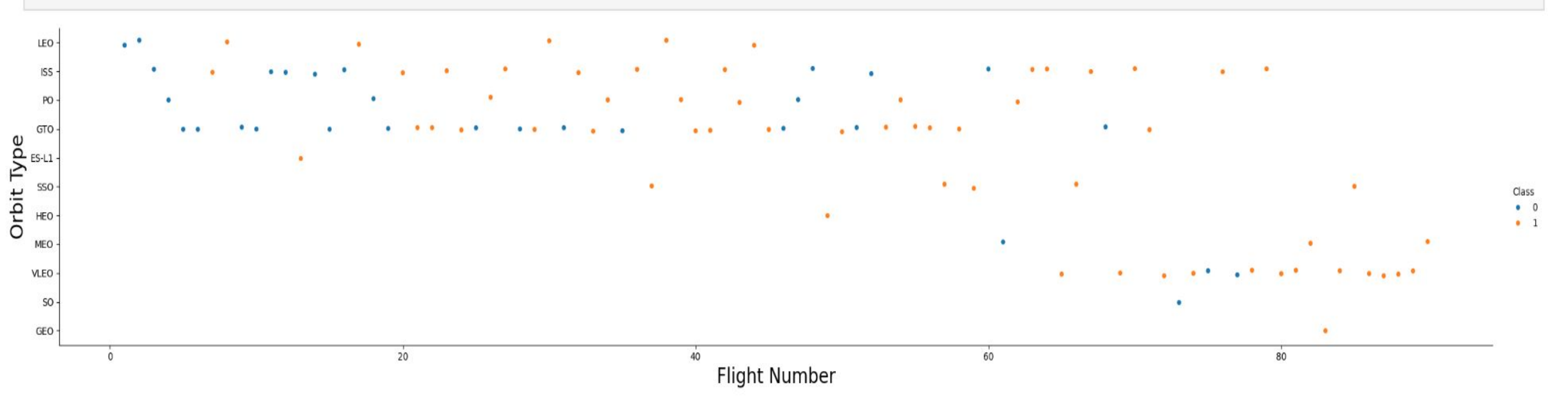
1. CCAFS had a mixture of success and failure for payload mass ranging from 400 to 7000 (approximately). However, they had large success for payload masses at 15000 approximately.
2. VAFB shows a steady progression of success after failed attempts at lighter payload masses.
3. KSC shows perfect success between payload ranges of 2000 to 5000 (approximately). However, failures were observed at 6000 payload mass (approximately).
4. Once again, CCAFS seems to be a research phase while KSC seems to be a standarizing operation based on the research. CCAFS was for iterative testing of different payload masses to see what works and what doesn't. They took this research and standarized it for KSC and VAFB, despite some issues prevailing.

Success Rate vs. Orbit Type

1. ES-L1, GEO, HEO, and SSO had perfect success rates.
2. VLEO was a close runnerup to the orbit types mentioned above.
3. LEO, MEO, PO, ISS, and GTO were similar in success rates (hovering between 0.7 to 0.5 approximately).
4. SO had zero success.
5. Some orbits may be more challenging than others. However more data is needed to understand.

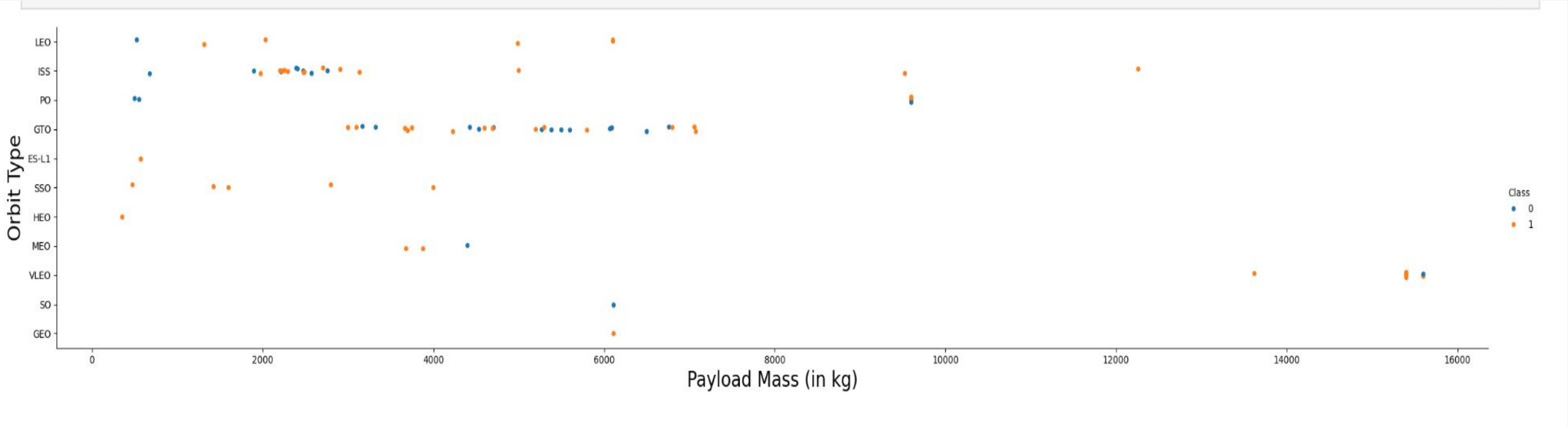


Flight Number vs. Orbit Type



1. Generally across all orbit types, the higher the flight number, the more likely a successful landing outcome is present.
2. GTO had the most failures present consistently, regardless of flight number (13).
3. SSO, HEO, MEO, VLEO, SO, and GEO had a low number of flights.
4. GEO had the fewest data points.
5. The lower flight numbers for LEO, ISS, PO, and GTO suggest a R&D phase that paid off for the most part (some struggles were still present for GTO). These processes learned seemed to be applied to SSO, HEO, MEO, VLEO, SO, and GEO, with success occuring.

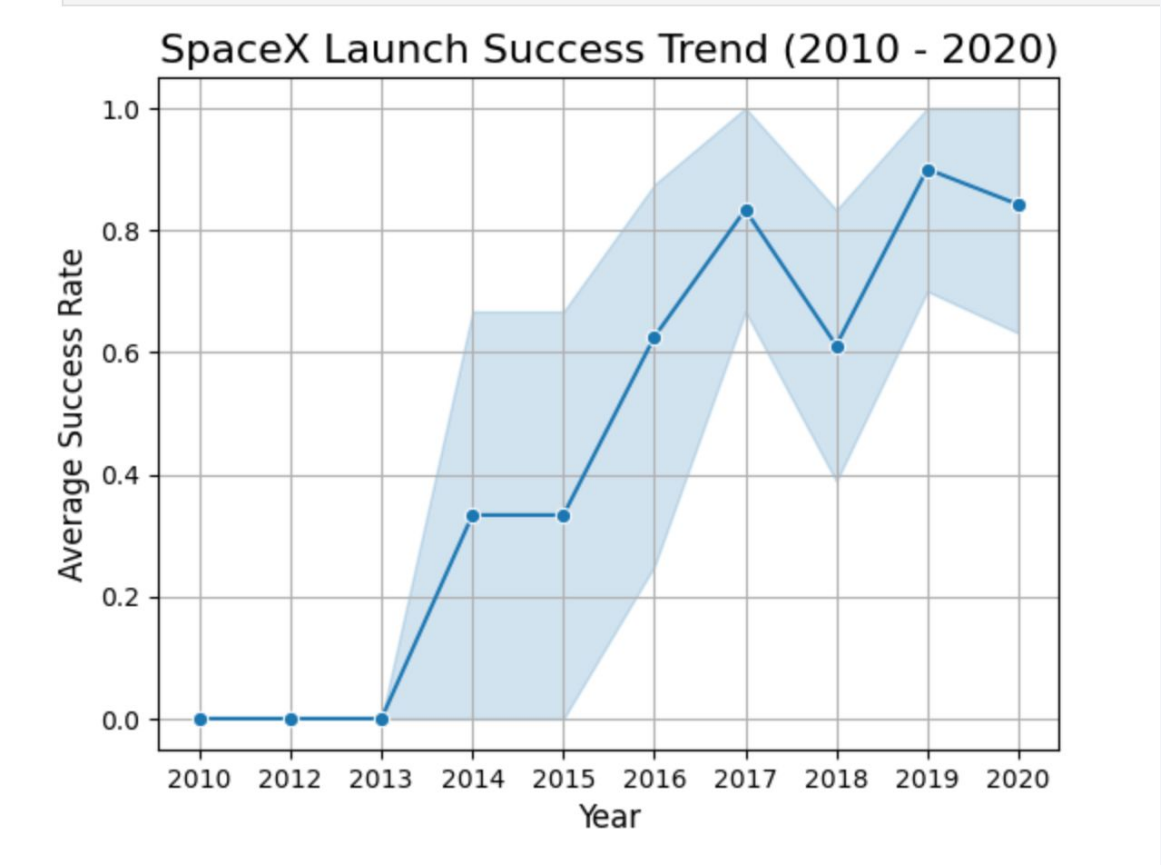
Payload vs. Orbit Type



1. GTO had a mix of success and failure with payload masses ranging from approximately 3000 to 7000.
2. SSO performed well on payload masses between approximately 800 and 4000, though with few data points.
3. ISS had a lot of success, with some failure, between payload masses of approximately 2000 and 3300.
4. VLEO had major success at high payload masses of approximately 13000 and 15000. However more data is needed.
5. ISS and GTO seem to be the R&D phase. What was learned was then applied for the other orbit types.

Launch Success Yearly Trend

1. Success was not present until after 2013.
2. Between 2013 and 2017, there was a major increase in success.
3. 2019 had the most success.
4. Slight decreases in success were noted between 2017 and 2018; 2019 and 2020.
5. Technology improvement, trial and error phases, and learning curves being overcome are indicative of this success trend.



All Launch Site Names

- Four unique launch sites:
  - CCAFS LC-40
  - VAFB SLC-4E
  - KSC LC-39A
  - CCAFS SLC-40

- **%sql** SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with `CCA`:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- **%sql** SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE'CCA%' LIMIT 5;

Total Payload Mass

- Total payload carried by boosters from NASA (CRS):
  - 45,596 kg

- **%sql** SELECT SUM("PAYLOAD_MASS__KG_") AS Total_Payload_Mass_NASA FROM SPACEXTABLE WHERE "Customer" = "NASA (CRS)";

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1:
  - 2928.4 kg

- **%sql** SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';

First Successful Ground Landing Date

- Dates of the first successful landing outcome on ground pad:
  - December 22, 2015

- **%sql** SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

PAYLOAD_MASS__KG_	Booster_Version
4696	F9 FT B1022
4600	F9 FT B1026
5300	F9 FT B1021.2
5200	F9 FT B1031.2

- **%sql** SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome"='Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000;

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes:

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- **%sql** SELECT "Mission_Outcome", COUNT(*) FROM SPACEXTABLE GROUP BY "Mission_Outcome";

Boosters Carried Maximum Payload

- Names of the booster which have carried the maximum payload mass (15,600 kg):

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_"=(SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);

2015 Launch Records

- Failed landing outcomes in drone ship, their booster versions, and launch site names for the year 2015:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- **%sql** SELECT substr(Date, 6, 2) AS Month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Failure (drone ship)' AND substr(Date, 0, 5) = '2015';

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Ranked count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

Landing_Outcome	Date	Outcome_Count
No attempt	2012-05-22	10
Success (drone ship)	2016-04-08	5
Failure (drone ship)	2015-01-10	5
Success (ground pad)	2015-12-22	3
Controlled (ocean)	2014-04-18	3
Uncontrolled (ocean)	2013-09-29	2
Failure (parachute)	2010-06-04	2
Precluded (drone ship)	2015-06-28	1

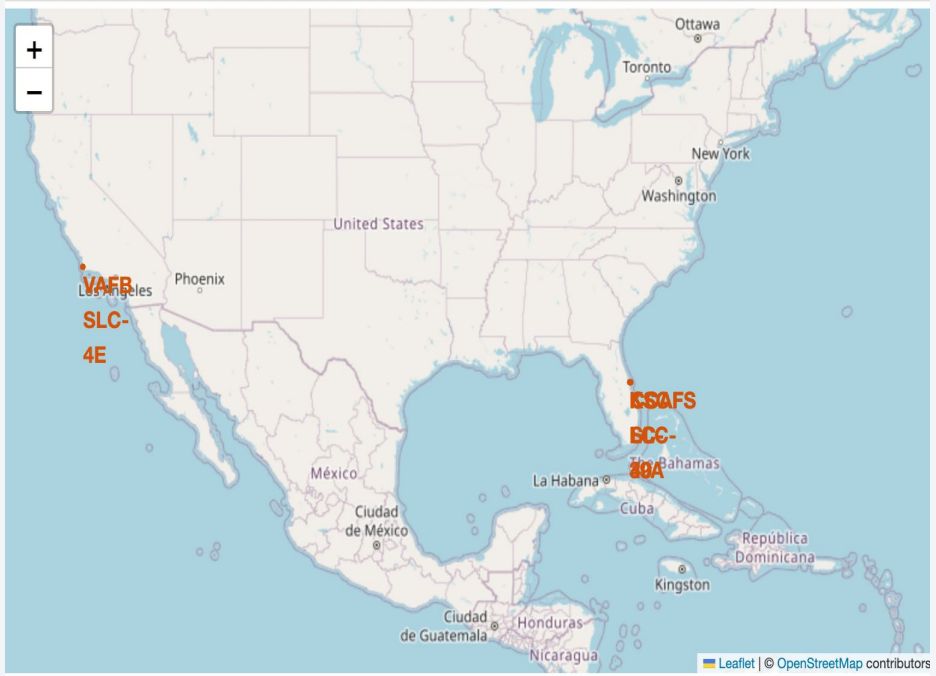
- %sql SELECT "Landing_Outcome", "Date", COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN "2010-06-04" AND "2017-03-20" GROUP BY "Landing_Outcome" ORDER BY "Outcome_Count" DESC;

Section 3

Launch Sites Proximities Analysis

Folium Map: Launch Sites

- Launch sites are located near the coastal regions of California and Florida to avoid any detrimental effects to human inhabited areas. In the event of failed launches/landings, the debris will fall into the ocean and not on a city.

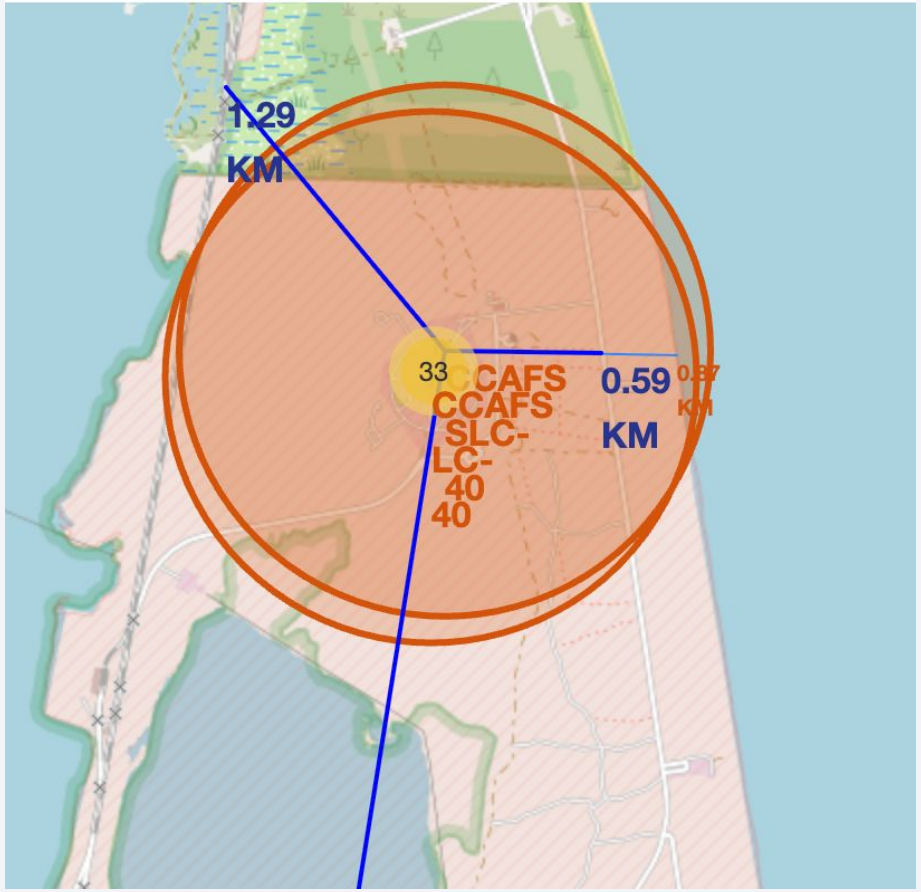


Folium Map: Launch Outcomes

- 56 total launch results:
  - CCAFS LC-40: 7 Success, 19 Failure
  - CCAFS SLC-40: 3 Success, 4 Failure
  - KSC LC-39A: 10 Success, 3 Failure
  - VAFB SLC-4E: 4 Success, 6 Failure

Folium Map: Proximities

- Roadway: 0.59 km
- Coast: 0.87 km
- Railway: 1.29 km
- City: 19.26 km

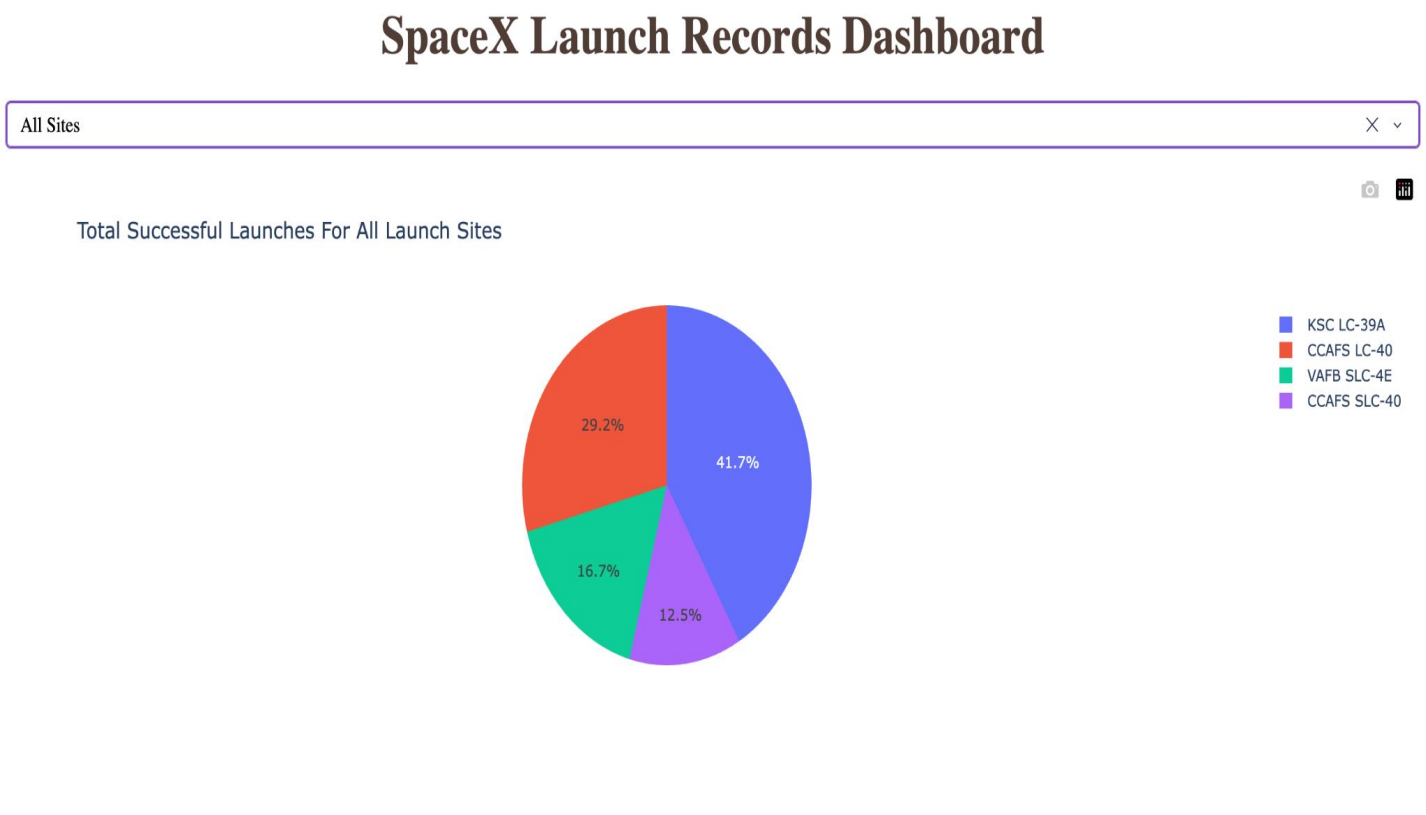


Section 4

Build a Dashboard with Plotly Dash

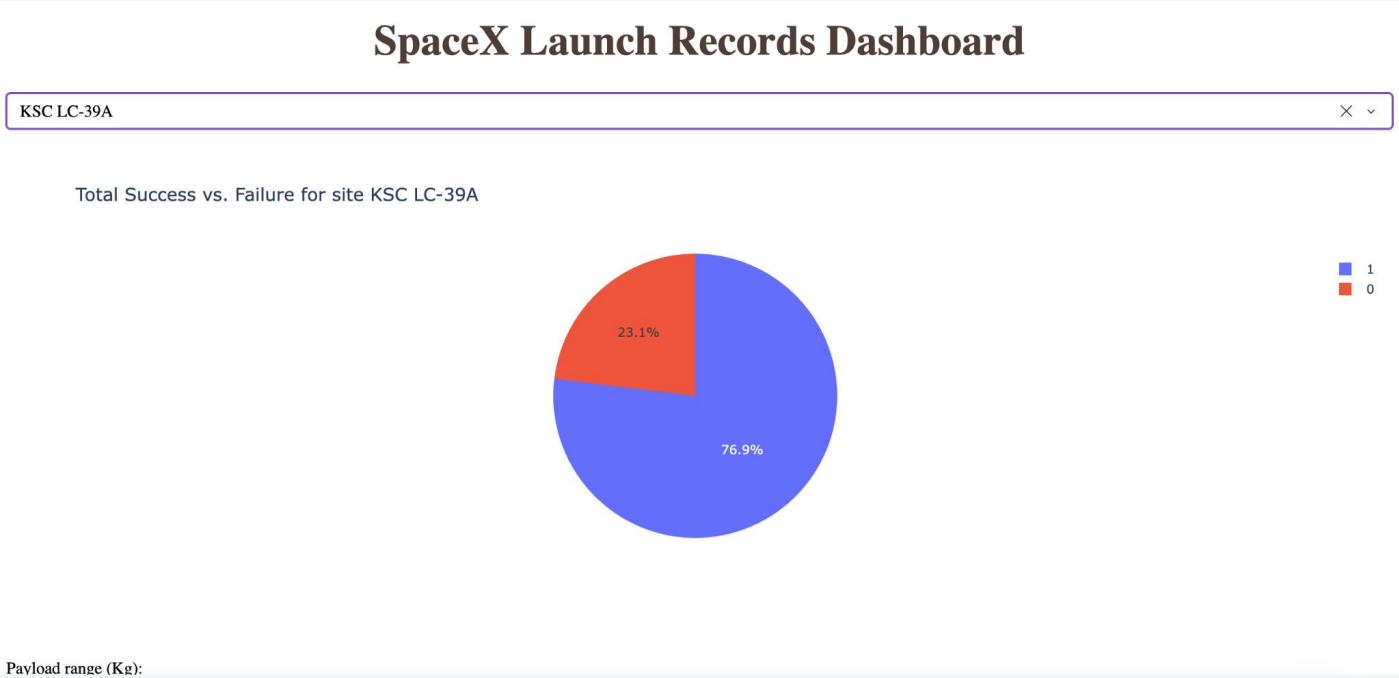
Dashboard: Launch Success (All Sites)

- KSC LC-39A (1st):
  - 10 success, 3 failure
- CCAFS LC-40 (2nd):
  - 19 success, 7 failure
- VAFB SLC-4E (3rd):
  - 6 success, 4 failure
- CCAFS SLC-40 (4th):
  - 4 success, 3 failure



Dashboard: Highest Launch Success Ratio

- KSC LC-39A had the highest success with a 10:3 ratio.



Dashboard: Payload vs. Launch Outcome (All Sites)

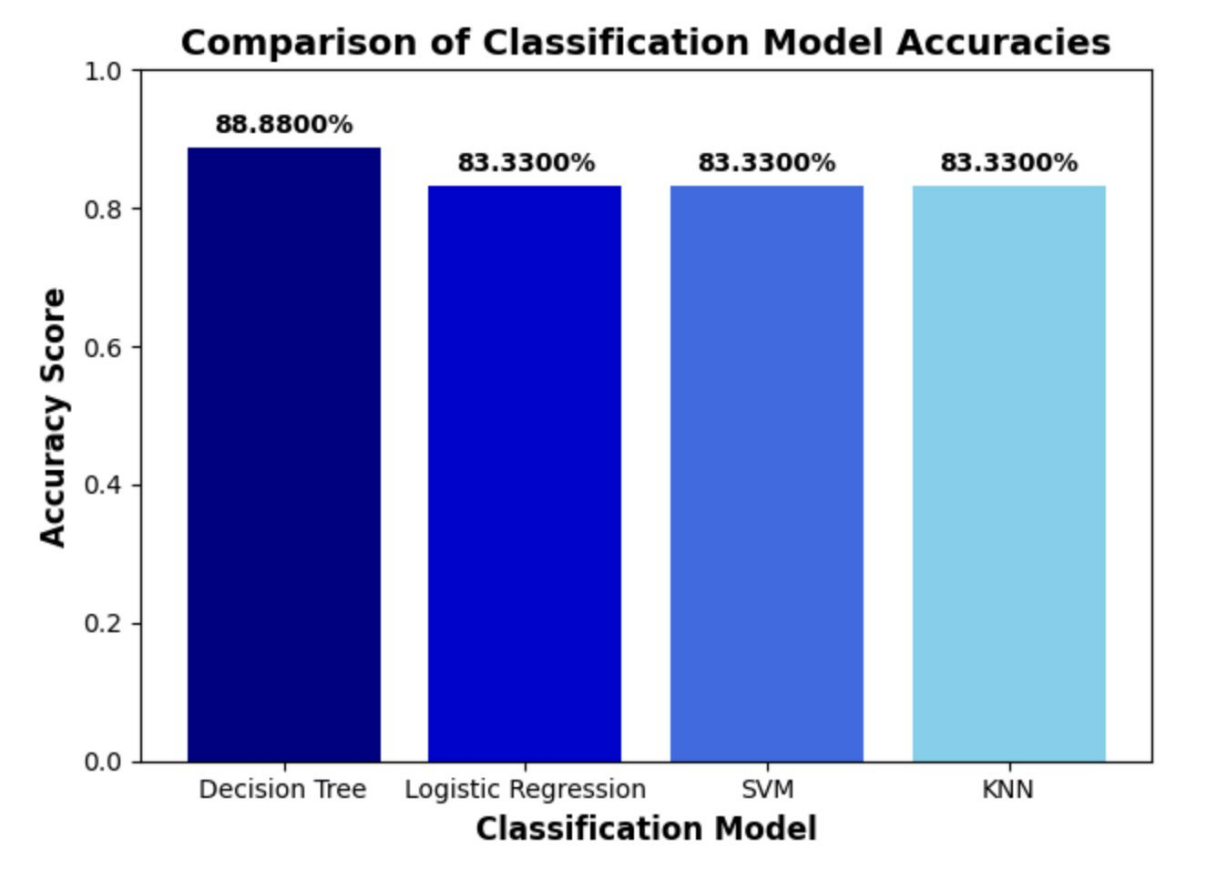
- The most successful paylaod range observed was between 2000 and 6000 kg:
  - Within this payload range, the booster version with the highest number of successful landings was FT
  - Within this payload range, the booster version with the least number of successful landings was v1.1
- Success Rate
  - B5: 1/1
  - FT: 13/20
  - B4: 6/11
  - V1.1: 1/15
  - V1.0: 0/3

Section 5

Predictive Analysis (Classification)

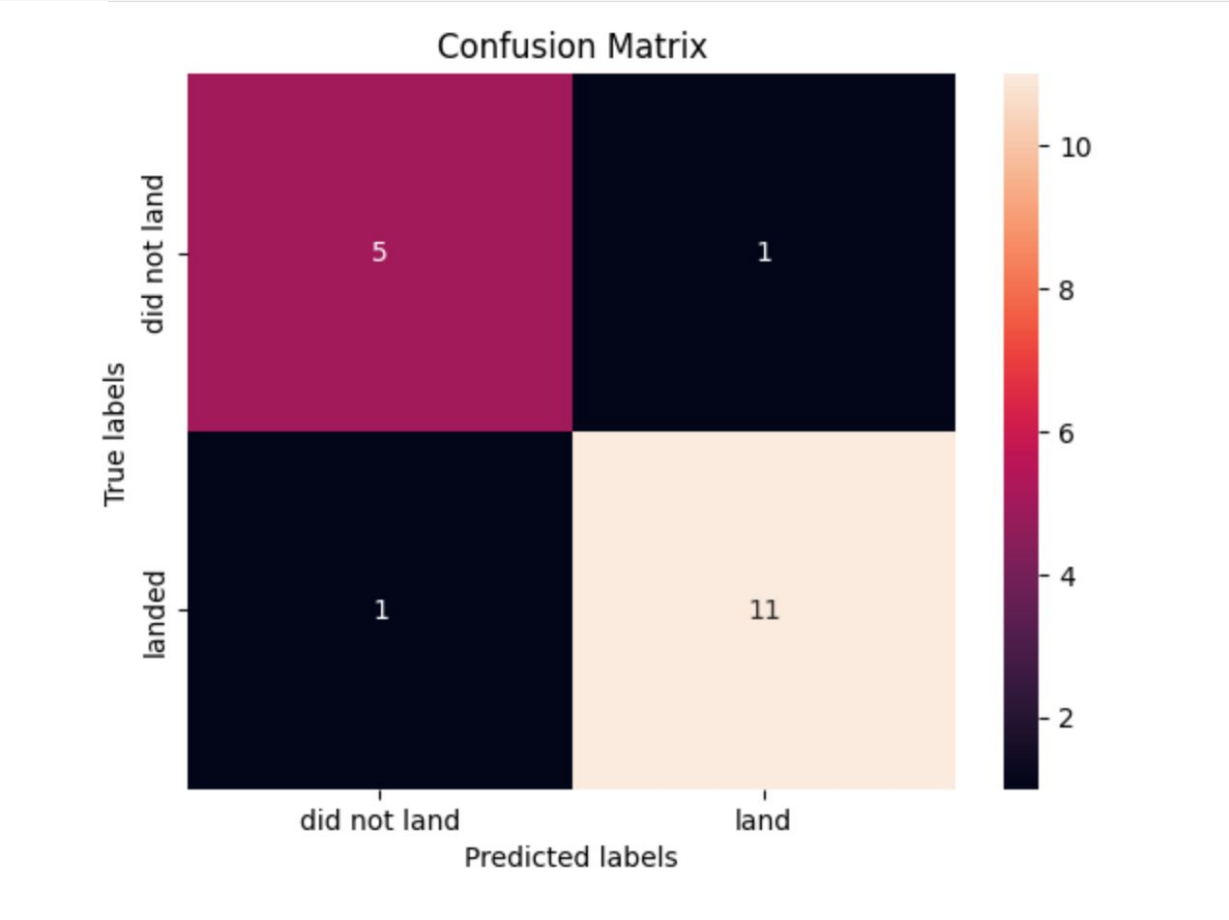
Classification Accuracy

- Decision Tree has the highest classification accuracy.



Confusion Matrix

- True Positives:
  - 11
- True Negatives:
  - 5
- False Positives:
  - 1
- False Negatives:
  - 1
- Total:
  - 16/18



Conclusions

- Success rates increased over time for all factors, demonstrating a clear learning curve. The initial success rates were up and down, which suggests that they were in a R&D phase that resulted in technological insights and advancements that benefited later success in landing outcomes.
- Orbit types ES-L1, GEO, HEO, and SSO had major success rates, however, this may have been once again after research and development phases. Specifically, orbit types LEO, ISS, PO, and GTO may have contributed to this in R&D.
- The most successful site was KSC LC-39A. It can be reasonably assumed that this site was for the operational phase, while site CCAFS LC-40 was for the R&D phase.
- The payload range of 2000-6000 kg (approximately) had the highest number of successful landing outcomes, revealing a “sweet spot” to be used for successful landing outcomes.
- The Decision Tree Classifier model had the highest testing accuracy with 88.88%. However, the other three models also had a high testing accuracy of 83.33%. This indicates that any of the models could be reasonably used and trusted.

Appendix

- Collected Data (finished versions)
  - From Spacex API: Lab #1
  - From Wikipedia Table: Lab #2
  - After Data Wrangling: Lab #3
  - After Data Visualization: Lab #5

Thank you!